Smart Civic Complaint System

Module-by-Module Progress & Technical Breakdown Report

Date: October 11, 2026
Git Branch: main (Commit 95a7d48)
Team: Smart Civic Engineering

OVERALL SYSTEM	BACKEND API	FRONTEND APP	LIVE ENDPOINTS
78%	80%	75%	20+
Total architecture completion	Express, Mongo, Multer & Auth	React 18, Vite & Tailwind/CSS	REST services tested & verified

Detailed Module Completion Breakdown

#	MODULE / FEATURE	BACKEND	FRONTEND	OVERALL	STATUS	IMPLEMENTATION DETAILS & REMAINING TASKS
1	**Authentication & RBAC**	95%	95%	**95%**	Completed	Multi-collection JWT auth (Citizen, Admin, Dept), Google OAuth 2.0 verification, bcrypt security, session timeout handling, and protected routes.
2	**Complaint Lifecycle**	90%	90%	**90%**	Completed	Issue submission, auto sequential IDs (SC-2026-XXXX), category auto-routing to dept, status transition pipeline, audit timeline, and 1-5★ ratings.
3	**Media & Photo Uploads**	90%	90%	**90%**	Completed	Multer disk storage engine, 5MB limits, file type validation (jpg/png/webp), and UI previews for complaint images and resolution inspection photos.
4	**In-App Notifications**	85%	85%	**85%**	Completed	Automated status-change alert triggers, /api/notifications feed, unread count, and mark-as-read endpoints connected to citizen notifications.
5	**Department Management**	80%	65%	**75%**	In Progress	Live computed complaint counts & SLA resolution rates API built; dept portal wired. *Remaining: Link admin Departments.jsx table & modals.*
6	**GIS Map & Hotspots**	60%	50%	**55%**	In Progress	Coordinates (lat/lng) persisted on complaints; Google Maps layout ready. *Remaining: Feed live MongoDB coordinates into ComplaintMap.jsx pins.*
7	**Admin User Management**	40%	50%	**45%**	In Progress	Citizen, Admin & Dept schemas exist; UI tables ready. *Remaining: Build /api/users to list all users across collections and toggle active/suspended status.*
8	**Analytics & Dashboards**	45%	45%	**45%**	In Progress	Dashboard KPI stats endpoint built; charts ready. *Remaining: MongoDB aggregation pipeline for monthly trends, category charts & ward heatmaps.*
9	**Settings & Help Desk**	30%	35%	**30%**	Pending	Setting and SupportTicket models seeded; frontend screens exist. *Remaining: Build*

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						`/api/settings` and `/api/support` controllers and routes.

Completed in Current Sprint

- Implemented full complaint CRUD with auto-ID & auto-routing.
- Built Multer file upload engine with 5MB validation.
- Created automated in-app notification triggers and API.
- Built department controller with real-time computed resolution rates.
- Replaced simulated submissions with live backend connections in citizen and department portals.

Next Sprint Priorities

- Build **Admin User Management API** (`/api/users`) to view, suspend, or delete accounts across collections.
- Build **Analytics Aggregation Pipeline** (`/api/analytics/dashboard`) for monthly trends and ward charts.
- Connect **Departments.jsx** and **ComplaintMap.jsx** to live backend data.